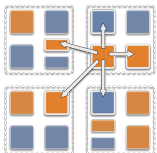


GPU Integration

- GPUs are supported in Charm4py via the Charm++ HAPI (Hybrid API) interface
- Allows asynchronous completion detection of GPU kernels via Charm4py futures, using the function
`charm.hapiAddCudaCallback(stream, future)`
- The runtime sets the future when all work in the stream before the callback is completed



Example

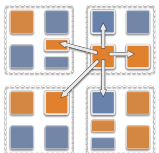
```
from charm4py import charm
import numba.cuda as cuda

def main(args):
    N = 1_000_000
    array_size = (N,)

    s = cuda.stream()
    stream_handle = s.handle.value
    . . .

    example_kernel[blocks_per_grid, threads_per_block, s](A_gpu, B_gpu)

    return_fut = charm.Future()
    charm.hapiAddCudaCallback(stream_handle, return_fut)
    return_fut.get() # wait on kernel completion
```



Future plan

- We plan to improve GPU support in Charm4Py by exposing the following Charm++ functionality
 - Portable support for GPUDirect RDMA
 - GPU-aware load balancing

